

IBVAP — Complete Production-Grade Feature Catalogue

Intelligent Border Video Analytics Platform

Purpose: This document lists the complete feature scope for a full-fledged IBVAP platform and explains what every feature does, what it consumes, what it produces, who uses it, what security or governance it requires, and whether it belongs to the current controlled prototype, a pilot, or the production target.

Important status convention:

Label	Meaning
Prototype	Can be demonstrated with controlled recorded video, simulated cameras, prepared metadata, or local services. It must be labelled as simulated where applicable.
Pilot	Requires a limited deployment with authorized live cameras, representative data, measured performance, and operator testing.
Production target	Required for a hardened multi-site deployment with security, resilience, governance, monitoring, backup, and operational approval.
Restricted	Requires separate policy, legal/operational approval, quality gates, access restrictions, human verification, and audit.
To be measured	No performance, accuracy, savings, or response-time result should be claimed until it is tested and documented.

IBVAP is an **assistive human-review platform**. It is not an autonomous threat-decision, identity, or intent-classification system.

1. Product-level capabilities

1.1 Software-defined CCTV upgrade

What it does: Adds intelligence around compatible existing IP cameras instead of requiring every camera to be replaced by a specialized smart camera.

Inputs: Existing IP camera streams, camera capabilities, site configuration, policies, and optional approved sensor metadata.

Outputs: Structured observations, rule events, incidents, alerts, evidence references, health states, and review records.

Users: Operators, supervisors, administrators, analysts, evidence custodians, and system integrators.

Status: Prototype concept and controlled replay are feasible now; production requires camera-compatibility testing and deployment approval.

1.2 Hybrid edge-plus-central processing

What it does: Places low-latency video processing and resilience functions close to cameras while centralizing incident intelligence, evidence governance, analysis, and command-center operations.

Edge responsibilities: Stream ingestion, frame sampling, quality checks, detection, temporary tracking, local rules, health checks, local queue, and prioritized synchronization.

Central responsibilities: Event normalization, deduplication, incident correlation, BLA, evidence control, audit, reporting, user permissions, and multi-site operations.

Status: Production architecture selected; only part of the prototype is currently implemented.

1.3 Human-in-the-loop decision support

What it does: Separates detection from review and consequential decision-making. AI and rules surface candidates; authorized personnel review the evidence and record an action.

Required safeguards: Review-required labels, reasoned actions, role checks, site scope, visible uncertainty, evidence-gap display, and audit records.

Status: Prototype frontend workflow exists, and protected prototype BLA review procedures are implemented and tested. Full production workflow is pending.

2. Identity, users, roles, and administration

2.1 User authentication

What it does: Authenticates operators and administrators through the platform's identity system.

Recommended production method: Institutional OIDC/OAuth with MFA for sensitive roles. The current managed project has a Manus OAuth scaffold.

Outputs: Authenticated session, user identity, role, allowed scope, session expiry, and audit event.

Security: Secure short-lived sessions, HTTP-only secure cookies where applicable, token rotation, expiry, and no long-lived tokens in browser storage.

Status: Authentication scaffold exists in the current project; institutional production identity integration is pending.

2.2 Role-based access control

What it does: Controls which functions a person may use.

Recommended roles: Camera operator, reviewer, supervisor, site administrator, system administrator, model administrator, evidence custodian, auditor, and integration service.

Current prototype roles: `user` and `admin`.

Security rule: Authorization must be enforced by backend procedures and APIs. Hiding a button in the frontend is not authorization.

Status: Basic role support exists; granular production roles are pending.

2.3 Site, camera, incident, and evidence scope

What it does: Limits a user to approved sites, cameras, incidents, and evidence objects.

Example: An operator assigned to Site North should not automatically access Site West evidence or configuration.

Status: Required production feature; not fully implemented in the current prototype.

2.4 User and role administration

What it does: Allows authorized administrators to provision, disable, assign, review, and revoke users and roles.

Controls: Approval workflow, audit records, least privilege, separation of duties, and periodic access review.

Status: Production target.

2.5 Configuration administration

What it does: Manages site settings, camera profiles, zones, schedules, rules, retention, alert policies, model deployments, and integrations.

Controls: Versioning, approval, change history, rollback, and administrator scope.

Status: Production target; current frontend contains an administration destination but not the complete backend.

3. Site and camera operations

3.1 Site registry

What it does: Creates a controlled record for each Border Out Post, checkpoint, border road sector, installation, or other surveillance site.

Data: Site ID, name, location, timezone, operational status, owning organization, network profile, retention policy, and assigned users.

Outputs: Site scope for cameras, rules, events, incidents, maps, and access control.

Status: Production target; prototype domain persistence is pending.

3.2 Camera registry

What it does: Registers each camera and links it to a site, location, capability profile, stream reference, and operational policy.

Data: Camera ID, site ID, vendor/model, stream protocol, resolution, frame rate, timezone, physical location, orientation, capabilities, health baseline, and secret reference.

Security: Store only a protected reference to credentials. Never expose passwords or raw stream secrets in the frontend or logs.

Status: Production target; current frontend simulates camera records.

3.3 Secure camera onboarding

What it does: Adds a camera through a guided workflow that validates connectivity and capabilities without exposing credentials.

Workflow: Select site → enter protected stream reference → discover or validate camera → test connection → inspect capability profile → assign zones and policy → approve camera

→ begin health monitoring.

Status: Production target; recorded-video simulation is the safe first prototype path.

3.4 RTSP/RTSPS ingestion

What it does: Receives video from compatible IP cameras using standard stream protocols.

Controls: Reconnection, buffering, timestamps, frame-rate control, stream isolation, health state, and secure credential handling.

Status: Prototype can use MP4 replay; live RTSP/RTSPS requires implementation and authorized testing.

3.5 ONVIF interoperability

What it does: Uses ONVIF discovery and supported profiles to identify camera capabilities and, where supported, exchange analytics metadata or events.

Important boundary: ONVIF support does not eliminate the need for RTSP/RTSPS and should not force the system to one vendor.

Status: Production target. ONVIF Profile M is the recommended reference for analytics metadata and event interoperability. ¹

3.6 Camera capability profiles

What it does: Records what each camera can actually provide, such as resolution, frame rate, PTZ, audio policy, analytics metadata, night capability, and stream profiles.

Why it matters: The platform must not assume that every camera has the same quality or features.

Status: Production target.

3.7 Stream session management

What it does: Creates authorized, short-lived browser playback sessions through HLS or WebRTC.

Security rule: Raw RTSP URLs must never be sent to the browser.

Status: Production target.

3.8 PTZ control, if permitted

What it does: Allows authorized users to control pan, tilt, and zoom on supported cameras.

Controls: Role restriction, site scope, command audit, rate limits, manual override, and operational approval.

Status: Optional production feature; not required for the first SIH prototype.

3.9 Camera health monitoring

What it does: Detects online, offline, frozen, stale, dark, blurred, obstructed, degraded, or repeatedly reconnecting camera states.

Inputs: Connection status, frame age, frame repetition, blur, brightness, glare, obstruction indicators, input FPS, and decoder health.

Outputs: Health state, degradation event, operator warning, incident context, and evidence-gap signal.

Key innovation: A camera that cannot see must not silently be interpreted as evidence that nothing happened.

Status: Health concept is represented in the prototype; real measurements and production monitoring are pending.

3.10 Camera coverage and blind-spot registry

What it does: Records which physical corridors, zones, roads, gates, and paths each camera covers and where coverage is missing.

Outputs: Coverage map, possible route gaps, unobserved corridors, and BLA evidence-gap context.

Status: Production target; central to the Border Shadow Map.

4. Edge video processing

4.1 Recorded-video simulator

What it does: Treats named MP4 clips as virtual cameras so the complete workflow can be demonstrated without unstable live equipment.

Inputs: MP4 files and a scenario manifest.

Outputs: Camera timestamps, frames, detections, rule events, evidence references, and repeatable test results.

Status: Required prototype feature; safe for SIH demonstration.

4.2 Frame decoding and sampling

What it does: Uses FFmpeg/OpenCV to decode streams, select frames at a controlled rate, resize them for inference, and preserve source/capture timestamps.

Why it matters: Every frame does not necessarily need to be analysed. Sampling controls CPU/GPU use and bandwidth.

Status: Production target; replay implementation is the recommended first step.

4.3 Timestamp normalization

What it does: Separates source time, capture time, receipt time, and central-ingestion time using UTC-based internal values.

Why it matters: BLA, incident correlation, evidence ordering, and offline synchronization depend on reliable time.

Status: Required across all stages.

4.4 Frame quality assessment

What it does: Measures blur, darkness, glare, freeze, repeated frames, obstruction, frame age, and usable visibility.

Outputs: Quality score, visibility state, degradation reason, and confidence context.

Safety: Poor quality should lead to lower confidence, insufficient evidence, or abstention, not a stronger claim.

Status: Production target; prototype may use controlled quality states.

4.5 Per-camera worker isolation

What it does: Runs camera pipelines so a failed stream or decoder does not crash all cameras at a site.

Controls: Supervisor process, bounded memory, restart policy, health heartbeat, and queue protection.

Status: Production target.

4.6 Bandwidth-aware event transmission

What it does: Sends prioritized metadata, key frames, and selected clips instead of attempting to transmit all raw video centrally under constrained links.

Policy: Critical events first, routine metadata next, bulk video only when permitted and available.

Status: Production target; important for remote sites.

5. Computer vision and AI features

5.1 Person detection

What it does: Finds person-shaped objects in frames and returns bounding boxes and model confidence.

Algorithm: Evaluated pretrained YOLO-family detector candidate.

Outputs: Class, bounding box, confidence, timestamp, camera, model version, and quality state.

Prototype training: No custom training required for the first controlled demonstration.

Production requirement: Measure precision, recall, mAP, latency, and performance by day/night/weather/quality condition.

5.2 Vehicle detection

What it does: Finds vehicles and returns bounding boxes, class labels, and model confidence.

Possible classes: Car, truck, bus, motorcycle, or other configured classes supported by the selected model.

Status: Prototype candidate and production target; class accuracy is to be measured.

5.3 Vehicle classification

What it does: Assigns a configured vehicle category to a detected vehicle.

Safety: Class output is probabilistic and should retain confidence and quality context.

Status: Prototype or pilot depending on data quality; production requires per-class evaluation.

5.4 Temporary multi-object tracking

What it does: Associates detections across frames within a camera and assigns temporary

IDs such as `cam-02-track-51` .

Recommended algorithm: ByteTrack-style association. 2

Outputs: Track path, entry/exit times, direction, dwell duration, object class, and evidence references.

Critical boundary: A temporary track ID is not a person identity and does not prove continuity across cameras.

5.5 Occlusion and re-entry context

What it does: Records when a temporary object becomes hidden, leaves the frame, or returns.

Why it matters: Helps the operator understand tracking limitations and prevents false claims of uninterrupted observation.

Status: Production target.

5.6 Face detection

What it does: Locates a face-shaped region and evaluates whether the image is usable.

Status: Restricted and not required for the first SIH prototype.

Controls: Policy authority, role restriction, quality threshold, reason logging, retention limit, audit, and human verification.

5.7 Face quality assessment

What it does: Determines whether a face crop is sufficiently clear for any approved downstream purpose.

Possible checks: Resolution, blur, pose, illumination, occlusion, and face visibility.

Safety: An unqualified face crop must be rejected or labelled insufficient rather than converted into a confident identity.

Status: Restricted production target.

5.8 Face matching or identification

What it does: Compares a quality-approved face against an authorized reference set when policy permits.

Status: Restricted, high-risk, and not required for the prototype.

Controls: Explicit authority, limited reference set, purpose logging, access scope, retention, bias/quality evaluation, human verification, audit, and approved operational procedure.

5.9 Plate localization

What it does: Locates a possible number plate in a vehicle crop.

Status: Optional restricted prototype/pilot feature; quality-gated.

5.10 ANPR/OCR

What it does: Converts a quality-approved plate crop into candidate text with confidence and quality state.

Safety: Low-quality frames must be rejected or shown as unreadable. OCR output is not automatically a confirmed vehicle identity.

Controls: Legal/policy authority, role restriction, search reason, retention, human verification, and audit.

Status: Restricted; not required for the first SIH prototype.

5.11 Detection confidence and abstention

What it does: Retains model confidence and quality context, allowing the system to abstain when evidence is insufficient.

Safety: Confidence is not the same as real-world correctness. Thresholds must be evaluated on representative authorized data.

Status: Required production behaviour.

6. Border-specific rule engine

The rule engine should use deterministic, versioned logic wherever a rule can be expressed through geometry, time, movement, or policy. This is easier to explain and audit than presenting every decision as an opaque AI result.

6.1 Virtual line crossing

What it does: Detects when a temporary track crosses a configured line in a configured direction.

Inputs: Track coordinates, line geometry, direction, timestamp, camera, and quality state.

Output: `line_crossing` event with line ID, direction, time, track ID, camera, and evidence reference.

Status: Required prototype and production feature.

6.2 Restricted polygon entry

What it does: Detects a transition from outside to inside a configured restricted polygon.

Inputs: Track point, polygon geometry, camera, zone policy, and timestamp.

Output: `restricted_zone_entry` event with zone ID, entry point, time, and evidence.

Status: Required prototype and production feature.

6.3 Wrong-direction movement

What it does: Detects when observed movement conflicts with the permitted direction for a line, road, corridor, or zone.

Outputs: `wrong_direction` event with configured direction and observed movement direction.

Status: Prototype target and production feature.

6.4 Dwell or loitering threshold

What it does: Detects when a temporary track remains in a configured zone longer than the approved threshold.

Outputs: `dwell_exceeded` event containing duration, threshold, zone, camera quality, and evidence.

Safety: “Dwell exceeded” is a rule event, not proof of suspicious intent.

Status: Prototype target and production feature.

6.5 Night-time movement schedule

What it does: Applies a site-specific timezone and schedule to classify movement during a restricted period.

Outputs: `night_movement` event with schedule version and timezone.

Status: Prototype target and production feature.

6.6 Camera-health rule

What it does: Emits an event when freeze, blur, darkness, obstruction, stale frame, or outage exceeds a threshold.

Output: `camera_degraded` event with measured indicator and threshold.

Status: Production target; prototype can simulate explicit degraded states.

6.7 Authorized exceptions

What it does: Prevents approved maintenance, patrol, convoy, or scheduled activity from being treated as an unexpected event when policy permits.

Controls: Time-limited exceptions, scope, approver, reason, audit, and automatic expiry.

Status: Production target.

6.8 Compound rules

What it does: Combines multiple independently recorded conditions into a higher-level event without hiding the components.

Example: Restricted-zone entry plus night schedule plus camera-quality state.

Safety: The system must show every contributing rule and must not present the compound event as an unexplained AI conclusion.

Status: Production target.

6.9 Rule versioning

What it does: Stores the exact geometry, threshold, schedule, exception, and policy version that generated an event.

Status: Production target and essential for auditability.

7. Event intelligence and incident management

7.1 Normalized event schema

What it does: Converts detector, tracker, quality, rule, and edge payloads into one common event contract.

Minimum fields: Event ID, idempotency key, gateway ID, site ID, camera ID, UTC occurrence time, receipt time, observation, model version, quality state, rule/version, evidence references, and schema version.

Status: Required before broad integration.

7.2 Event deduplication

What it does: Compresses repeated frame-level signals from the same track/rule/time window into one meaningful event.

Inputs: Source event IDs, track ID, rule ID, camera, time window, and idempotency key.

Outputs: Deduplicated event and deduplication explanation.

Status: Required production feature; not yet fully implemented.

7.3 Incident correlation

What it does: Groups related events from one or more cameras into one reviewable incident timeline.

Outputs: Incident ID, contributing events, direct observations, inferences, evidence gaps, priority, status, and review history.

Status: Production target; current frontend shows the concept.

7.4 Priority scoring

What it does: Assigns a configurable review priority using deterministic policy weights and contextual conditions.

Safety: Priority is not a final threat classification or criminality judgment.

Status: Prototype presentation concept; production policy engine pending.

7.5 Alert queue

What it does: Presents prioritized incident candidates to the correct authorized audience.

Filters: Site, camera, severity/priority, status, time, event type, evidence state, and assignment.

Status: Frontend prototype exists; backend integration is pending.

7.6 Alert acknowledgement

What it does: Lets an authorized operator confirm that an alert has been received.

Outputs: Updated status, actor, timestamp, optional note, and audit record.

Status: Simulated in the frontend; protected backend workflow foundation exists for BLA patterns.

7.7 Incident review actions

Possible actions: Start review, confirm review outcome, dismiss, escalate, resolve, and close.

Controls: Valid state transitions, role authorization, site scope, mandatory reason for consequential actions, idempotency, and audit.

Status: Protected prototype BLA review procedure is implemented and tested; general incident workflow remains pending.

7.8 Incident state machine

Plain Text

```
Detected
-> Normalized
-> Deduplicated
-> Correlated
-> Prioritized
-> Alerted
-> Acknowledged
-> Under Review
-> Confirmed / Dismissed / Escalated
-> Resolved
-> Closed
```

Every transition must record the actor, time, reason, prior state, new state, and related evidence/context.

7.9 Explainable alert summary

What it does: Explains why the incident exists in plain language.

Required content: Camera, time, object/rule, direct observation, inferred context, evidence quality, gaps, model/rule version, and required human action.

Example: “Possible repeated checkpoint movement pattern detected — human review required.”

Status: Prototype design exists; backend data integration pending.

8. Behavioral Link Analysis and Border Shadow Map

8.1 Recent-sightings ledger

What it does: Keeps a short-lived structured memory of recent observations for approved analysis.

Fields: Timestamp, site, camera, zone, object class, temporary track ID, direction, quality/confidence, evidence references, source model, source rule, and retention/expiry state.

Status: Prototype workflow service exists in memory; durable production ledger is pending.

8.2 Camera topology graph

What it does: Records which cameras or zones are physically or operationally related and what route relationships are permitted for analysis.

Outputs: Allowed camera pairs/routes, expected travel windows, direction context, and coverage gaps.

Status: Production target.

8.3 Time-window correlation

What it does: Checks whether sightings occur within a configured time window that makes a possible route relationship worth reviewing.

Safety: Time compatibility is evidence for a possible pattern, not proof of identity or continuity.

Status: Prototype target and production feature.

8.4 Possible perimeter-reconnaissance pattern

What it does: Surfaces a possible repeated or unusual sequence through configured perimeter cameras.

Required output: Supporting sightings, route assumptions, quality states, evidence gaps,

and “human review required” label.

Status: Prototype scenario and production target.

8.5 Possible shuttle-loop or rapid-turnaround pattern

What it does: Surfaces repeated appearance near a checkpoint within a policy window with a direction-change or turnaround context.

Required output: Direct observations, time window, route context, uncertainty, and evidence gap.

Status: Prototype scenario and production target.

8.6 Cross-camera possible-link explanation

What it does: Shows why two observations may be related based on time, topology, direction, class, or configurable appearance cues.

Safety: Must use “possible link” or “possible pattern,” not “confirmed same person.”

Status: Production target.

8.7 Evidence-gap evaluator

What it does: Identifies where continuity cannot be verified because a camera was unavailable, the route was not covered, the image was unusable, or the object was not observed.

Outputs: Gap state, affected corridor, last direct sighting, next direct sighting if any, cause, duration, and confidence limitation.

Status: Core innovation and production target.

8.8 Border Shadow Map view

What it does: Visualizes direct sightings, possible links, camera topology, blind spots, degraded coverage, and unobserved corridors on a map or operational diagram.

Users: Operators, analysts, supervisors, and auditors.

Status: Prototype concept is represented in the frontend; full GIS and backend data layer are pending.

8.9 BLA review authorization

What it does: Ensures only authenticated and appropriately authorized users can review or make decisions about BLA patterns.

Current prototype implementation: `bla.list` , `bla.get` , and `bla.review` protected tRPC procedures; operators can acknowledge/start review, while current final decisions are restricted to `admin` because the current user model only has `user` and `admin` roles.

Current tests: Unauthenticated access is rejected; operator acknowledgment/review works; operator final confirmation is rejected; administrator confirmation creates an audit record.

8.10 BLA retention and privacy controls

What it does: Limits how long recent sightings and possible links remain available and restricts access by site, purpose, and role.

Status: Production target.

9. Evidence management

9.1 Key-frame capture

What it does: Stores a selected frame associated with an event or incident.

Metadata: Evidence ID, object key, camera, time, event, hash, quality state, source model/rule, and retention state.

Status: Prototype reference can be demonstrated; secure storage integration is pending.

9.2 Short event clip

What it does: Preserves a controlled clip before and after a selected event where policy permits.

Controls: Retention, size limits, hash, access grant, export approval, and audit.

Status: Production target.

9.3 Evidence hash and integrity metadata

What it does: Records a cryptographic hash and provenance metadata to detect alteration.

Status: Production target; a prototype may show hash fields without claiming legal chain-of-custody certification.

9.4 Evidence metadata catalogue

What it does: Stores searchable metadata without placing large video bytes in normal database rows.

Storage rule: Object bytes go to private S3-compatible storage; the database stores references and metadata.

Status: Production target.

9.5 Short-lived evidence access grant

What it does: Issues an authorized, limited-time viewing reference rather than a public object URL.

Inputs: Evidence ID, user scope, purpose, and reason where required.

Outputs: Signed limited URL/session, expiry, and audit ID.

Status: Production target.

9.6 Controlled evidence export

What it does: Creates an export task for approved evidence and records purpose, requester, approver, objects, time, and delivery status.

Status: Production target; restricted by role and scope.

9.7 Retention and deletion policy

What it does: Automatically handles retention periods, legal holds where applicable, expiry, and deletion records.

Controls: Policy version, approval, audit, protection from unauthorized deletion, and storage monitoring.

Status: Production target.

9.8 Evidence access audit

What it does: Records every view, search, grant, export, denial, and policy action.

Status: Production target.

10. Command-center frontend features

10.1 Operations overview

What it does: Gives the operator a concise mission and system picture.

Displays: Monitoring posture, review queue, priority alerts, camera availability, open investigations, edge synchronization, immediate attention, and limitations.

Status: Implemented as a high-end prototype interface with simulated data clearly labelled.

10.2 Camera workspace

What it does: Lists cameras, health states, recent events, selected stream context, zones, and authorized playback.

Status: Prototype camera selection exists; real playback and backend data are pending.

10.3 Incident workspace

What it does: Shows alert queue, incident detail, timeline, evidence states, contributing events, possible links, gaps, reviewer actions, and audit sequence.

Status: Prototype interaction exists; full API integration is pending.

10.4 BLA workspace

What it does: Shows possible behaviour patterns, contributing sightings, topology, uncertainty, evidence gaps, and human-review controls.

Status: Frontend concept plus protected backend review workflow; complete persistence and live data are pending.

10.5 Map and site view

What it does: Displays sites, cameras, zones, routes, incidents, camera coverage, and Shadow Map gaps.

Status: Prototype concept; production GIS integration is pending.

10.6 Evidence workspace

What it does: Lists evidence metadata, quality, hashes, access state, review relationship, retention state, and authorized viewing/export controls.

Status: Prototype concept; secure evidence service is pending.

10.7 Health workspace

What it does: Shows camera availability, degradation, edge CPU/GPU/RAM/disk/temperature, queue depth, synchronization errors, and service state.

Status: Frontend simulation exists; real metrics are pending.

10.8 Metrics workspace

What it does: Shows measured operational indicators such as event volume, duplicate suppression, acknowledgement delay, queue recovery, camera availability, and model latency.

Safety: Simulated metrics must be labelled; no accuracy or impact claim is valid until measured.

Status: Prototype activity chart exists and is labelled as simulated scenario activity.

10.9 Administration workspace

What it does: Provides authorized configuration for users, sites, cameras, zones, rules, retention, model deployments, integrations, and system policy.

Status: Production target.

10.10 Responsive and accessible interface

What it does: Keeps the command center usable across desktop, tablet, and limited-screen contexts.

Requirements: Keyboard access, visible focus, semantic headings, screen-reader labels, non-colour status indicators, contrast, reduced-motion support, readable typography, error messages, loading states, and empty states.

Status: Accessibility direction is part of the frontend design; full audit is pending.

10.11 Notifications and real-time updates

What it does: Pushes authorized alert, incident, health, and synchronization changes to the dashboard.

Recommended transport: Authenticated WebSockets with typed event envelopes.

Status: Prototype UI state exists; production WebSocket backend is pending.

11. Offline-edge resilience

11.1 Edge durable queue

What it does: Stores event metadata and evidence references locally when the central connection is unavailable.

Recommended storage: SQLite at the edge.

Queue fields: Event ID, idempotency key, gateway/site/camera, event payload, evidence references, sync state, attempts, timestamps, and last error.

Status: Frontend offline simulation exists; durable queue implementation is pending.

11.2 Online/offline state machine

What it does: Clearly displays whether the edge gateway is connected, degraded, offline, synchronizing, or recovered.

Status: Simulated in the current UI; actual edge state service is pending.

11.3 Store-and-forward synchronization

What it does: Retries queued events after connectivity returns.

Controls: Bounded backoff, ordering policy, priority, retry limits, error classification, and audit of sync outcomes.

Status: Production target.

11.4 Idempotent central ingestion

What it does: Accepts a queued event once and rejects duplicate retries safely using a stable idempotency key.

Status: Required production feature; current BLA review service does not yet implement central event ingestion.

11.5 Evidence synchronization policy

What it does: Decides whether to synchronize metadata, key frames, clips, or full media based on priority, bandwidth, storage, and policy.

Status: Production target.

11.6 Restart and recovery handling

What it does: Preserves queue state across process or machine restart and resumes synchronization without data loss or duplication.

Status: Production target and mandatory resilience test.

11.7 Queue-full and storage-pressure handling

What it does: Responds safely when local storage is near capacity.

Controls: Priority preservation, retention policy, operator warning, backpressure, and no silent deletion of critical records.

Status: Production target.

12. Security, privacy, and governance

12.1 Backend authorization

Every protected procedure must check authentication, role, site scope, camera scope, incident scope, and evidence scope as applicable. The frontend must not be treated as a security boundary.

12.2 API validation

Use strict schemas, enumerations, size limits, safe errors, rate limits, and idempotency. Invalid or unauthorized payloads must fail safely.

12.3 Session and CSRF protection

Use secure session handling, expiry, rotation, HTTP-only cookies where applicable, and CSRF protection for cookie-authenticated mutations.

12.4 Secret management

Keep camera passwords, tokens, and API keys in a secret manager or injected environment. Never commit them, display them, or write them to logs.

12.5 Network segmentation

Separate camera networks, edge gateways, central services, administration, and external integrations. Use TLS and protected service identity such as mTLS where practical.

12.6 Data-at-rest protection

Encrypt databases, evidence storage, backups, and local edge volumes according to the deployment environment and key-management policy.

12.7 Audit logging

Record authentication, authorization denial, searches, evidence access, exports, rule changes, model changes, review actions, sync outcomes, and retention actions.

12.8 Supply-chain security

Pin dependencies, scan for vulnerabilities, review code, protect build pipelines, and maintain reproducible or signed build controls where required.

12.9 Model integrity

Store model name, version, artifact hash, dataset version, intended operating conditions, thresholds, approval status, deployment scope, and rollback target.

12.10 Privacy minimization

Collect and retain only what the approved purpose requires. Restrict sensitive identity data, preserve uncertainty, and apply retention and access policies.

12.11 Biometric safeguards

Face and plate capabilities are restricted features. They require policy authority, reason logging, quality rejection, role restrictions, retention limits, audit, and human verification. Face detection is not face identification; OCR output is not automatically confirmed identity.

13. Model lifecycle and AI governance

13.1 Model registry

What it does: Stores approved, candidate, shadow, and retired models with metadata and deployment scope.

13.2 Dataset registry

What it does: Stores dataset version, source authorization, labels, split strategy, annotation quality, and permitted use.

13.3 Model card

What it does: Documents intended use, limitations, data conditions, known failure modes, threshold, evaluation metrics, and operator guidance.

13.4 Shadow mode

What it does: Runs a candidate model beside the approved model without creating official alerts, enabling safe comparison.

13.5 Threshold management

What it does: Tracks thresholds by model, camera, class, quality condition, and policy version.

13.6 Rollback

What it does: Reverts an alert-producing model or rule configuration to an approved prior version.

13.7 Drift and quality monitoring

What it does: Watches for changes in scene, lighting, camera angle, weather, compression, object distribution, false-alert rate, abstentions, and operator feedback.

13.8 Retraining workflow

What it does: Supports authorized data collection, annotation, QA, versioning, scene/time split, fine-tuning, validation, held-out testing, model-card generation, shadow mode, approval, and rollback.

13.9 Human feedback loop

What it does: Uses reviewed outcomes to identify false alerts, missed events, quality problems, rule changes, and future training priorities.

Safety: Operator feedback must not automatically retrain or deploy a model without controlled validation and approval.

14. Integrations and interoperability

14.1 Command-and-control adapter

What it does: Sends approved incidents, locations, evidence references, and status to an existing command-and-control system.

Controls: Adapter boundary, schema mapping, authentication, retries, idempotency, scope, and audit.

Status: Production target; use a controlled adapter stub for the prototype if needed.

14.2 VMS adapter

What it does: Integrates with an existing video-management system where required.

Controls: No vendor lock-in, protected sessions, permission mapping, and audit.

Status: Optional integration; not required for first prototype.

14.3 Notification adapters

What it does: Sends approved notifications through configured channels such as command-center updates or institutional notifications.

Controls: Role, severity, escalation policy, suppression, acknowledgement, and audit.

14.4 GIS integration

What it does: Connects approved maps, site geometry, zones, corridors, and operational layers.

Status: Prototype can use a non-commercial map approach; production GIS source must be approved.

14.5 ONVIF adapter

What it does: Discovers compatible cameras and exchanges supported metadata/events.

Status: Production target.

14.6 External sensor adapter

What it does: Provides a future boundary for approved radar, thermal, seismic, access-control, or other sensor data.

Safety: External sensor signals must remain source-labelled and must not be merged into certainty without validation.

Status: Future production extension.

15. Observability, operations, and reliability

15.1 Camera monitoring

Track online/offline state, frame age, reconnects, blur, freeze, darkness, obstruction, input FPS, and stream errors.

15.2 Edge monitoring

Track CPU, GPU, RAM, disk, temperature, queue depth, synchronization retries, service health, and local storage pressure.

15.3 AI monitoring

Track inference duration, frames analysed, model version, threshold, abstentions, errors, and alert contribution.

15.4 Incident monitoring

Track event volume, duplicate suppression, priority distribution, acknowledgement delay, review age, escalation age, and unresolved backlog.

15.5 Evidence monitoring

Track storage volume, access events, exports, retention actions, hash failures, and backup status.

15.6 Security monitoring

Track failed logins, denied permissions, unusual searches, rule/model changes, exports, rate limits, and service identity failures.

15.7 Central logging and tracing

Use structured logs with correlation IDs, source service, site, camera, event, and request context. Do not log secrets or unnecessary sensitive image data.

15.8 Backup and recovery

Back up configuration, databases, audit records, and evidence metadata according to policy. Test restore procedures rather than only creating backups.

15.9 Operational runbooks

Document camera onboarding, stream failure, edge outage, queue recovery, evidence export, model rollback, incident escalation, backup restore, and security incident response.

16. API and real-time interface inventory

The following is the recommended production API surface. It is a target contract, not a statement that every route is currently implemented.

API	Purpose	Scope
GET /api/v1/sites	List authorized sites.	Scoped user.
POST /api/v1/sites	Create a site.	Administrator.
GET /api/v1/cameras	List cameras with filters and health.	Operator/site scope.
POST /api/v1/cameras	Register a camera.	Administrator.
GET /api/v1/cameras/{id}/stream-session	Obtain a short-lived HLS/WebRTC playback session.	Authorized viewer.
GET /api/v1/zones	List zone and virtual-fence definitions.	Site/camera scope.
POST /api/v1/zones	Create or version a zone.	Supervisor/administrator.

POST /api/v1/edge/events	Receive normalized, idempotent edge events.	Authenticated edge gateway.
GET /api/v1/events	Search normalized events.	Operator/analyst scope.
GET /api/v1/alerts	List prioritized alerts.	Operator scope.
POST /api/v1/alerts/{id}/acknowledge	Acknowledge an alert.	Operator/supervisor policy.
GET /api/v1/incidents/{id}	Retrieve explainable incident detail.	Scoped user.
POST /api/v1/incidents/{id}/review-actions	Confirm, dismiss, escalate, resolve, or close.	Role and scope policy.
GET /api/v1/behavior-patterns	List BLA outputs.	Analyst/operator scope.
GET /api/v1/behavior-patterns/{id}	Explain BLA evidence, links, and gaps.	Scoped user.
GET /api/v1/evidence/{id}	Retrieve evidence metadata.	Evidence scope.
POST /api/v1/evidence/{id}/access-grant	Issue a short-lived access grant.	Evidence custodian/scope.
POST /api/v1/evidence/exports	Create a controlled export task.	Supervisor/auditor policy.
GET /api/v1/health/summary	Retrieve camera/edge/platform health.	Operator/site scope.
GET /api/v1/metrics	Retrieve measured or explicitly simulated metrics.	Analyst/supervisor scope.
WS /api/v1/ws/operations	Push authorized alert, incident, health, and sync updates.	Authenticated scoped client.

The current managed project uses typed tRPC procedures. The target FastAPI/OpenAPI interface can be implemented later behind an adapter boundary, but the data and authorization semantics should remain consistent.

17. Production data entities

The production data model should include at least the following entities:

Entity	Purpose
User	Identity, role, status, and last sign-in.
Role/Permission	Fine-grained authorization.
Site	Physical and operational site record.
Camera	Camera capability, location, stream reference, and health.
CameraHealthSample	Time-series health and quality signals.
Zone	Versioned lines, polygons, schedules, and policies.
CameraTopologyEdge	Approved possible route relationship and expected travel window.
ModelVersion	Model artifact, version, hash, thresholds, and approval.
DatasetVersion	Data source, labels, split, authorization, and QA.
Detection	Per-frame object observation.
Track	Temporary per-camera temporal association.
Event	Normalized rule or health event.
Alert	Routed operator-facing alert.
Incident	Correlated review case.
BehaviorPattern	BLA candidate with observations, inference, gaps, and review status.
EvidenceObject	Private object-storage reference and integrity metadata.
EvidenceAccessGrant	Short-lived viewing authorization.
EvidenceExport	Controlled export request and status.
ReviewAction	Human action, rationale, actor, and state transition.
AuditEntry	Append-oriented security and workflow audit record.

EdgeGateway	Gateway identity, site scope, health, and deployment version.
EdgeQueueItem	Local or central synchronization state.
NotificationDelivery	Delivery, retry, acknowledgement, and failure state.
RetentionPolicy	Data class, period, legal hold, and deletion rules.
IntegrationEndpoint	C2/VMS/GIS/notification adapter configuration.

The current Drizzle schema contains only the core `users` table. These domain entities remain future implementation work and must be introduced through schema-first migrations.

18. Testing and acceptance features

18.1 Unit tests

Test geometry, virtual fences, direction, dwell, schedules, priority rules, BLA conditions, gap logic, state transitions, authorization helpers, and idempotency.

18.2 API integration tests

Test authentication, authorization, validation, persistence, error codes, site scope, evidence scope, review actions, and audit records.

18.3 Frontend tests

Test route navigation, filters, review controls, loading states, empty states, errors, keyboard access, and non-colour status cues.

18.4 End-to-end vertical slice

Test:

Plain Text

Recorded clip

- > virtual camera
- > quality check
- > detection
- > temporary track
- > virtual-fence/rule event
- > deduplication
- > incident
- > BLA context
- > evidence reference
- > human review
- > audit entry

18.5 Computer-vision replay tests

Use named clips and a scenario manifest with expected objects, zones, events, possible links, gaps, and review outcomes.

18.6 Resilience tests

Test network loss, process restart, queue full, duplicate synchronization, invalid event, camera failure, central downtime, and recovery.

18.7 Security tests

Test RBAC, site scope, evidence scope, secrets, API abuse, export authorization, CSRF, dependency scanning, and audit integrity.

18.8 Performance tests

Measure inference latency, stream count, database query load, queue growth, central throughput, synchronization delay, and resource utilization on defined hardware.

18.9 UAT/operator comprehension

Verify that operators can interpret direct observations, inferences, evidence gaps, camera-health states, and review actions without treating a possible BLA pattern as a confirmed fact.

19. Prototype scope versus full product

Area	SIH prototype	Production-grade platform
Input	Controlled MP4 clips and named virtual cameras.	Governed live multi-site IP cameras.
AI	Pretrained detector candidate.	Evaluated, versioned, monitored, shadow-tested models.
Tracking	Temporary per-camera IDs.	Capacity-tested, monitored tracking pipeline.
Rules	One or more virtual-fence, direction, dwell, or night scenarios.	Versioned multi-site policy engine with exceptions and audit.
BLA	Prepared pattern scenarios and visible gaps.	Durable recent-sightings ledger, topology, retention, scope, and measured usefulness.
Evidence	Controlled references or local/S3-compatible prototype.	Private storage, hashes, access grants, retention, export governance, and backup.
Review	Protected prototype workflow.	Full role/scope-based incident and BLA review with durable audit.
Offline	Explicit UI simulation.	Real edge queue, restart recovery, retry, idempotency, and failure testing.
Face/ANPR	Omit or show restricted placeholder.	Only after separate authority, quality evaluation, privacy controls, and human verification.
Metrics	Simulated scenario data clearly labelled.	Field-measured operational and model metrics.
Deployment	Developer laptop or controlled demo environment.	Hardened edge gateways and controlled central services.

20. Recommended implementation order

1. Stabilize repository, contracts, environment, and secret policy.
2. Define sites, cameras, roles, zones, event, incident, evidence, BLA, audit, and edge-queue schemas.
3. Add durable prototype persistence and protected backend procedures.
4. Complete camera and site registry.
5. Add recorded-video simulator and scenario manifest.
6. Add FFmpeg/OpenCV frame processing and quality states.
7. Add detector adapter and temporary tracking adapter.
8. Add virtual-fence, direction, dwell, and night rules.
9. Add normalized events, deduplication, priority, and incident correlation.
10. Add evidence references, hashes, access grants, and audit.
11. Replace frontend-only review state with protected backend procedures.
12. Add durable BLA recent-sightings ledger, topology, possible patterns, and Border Shadow Map.
13. Add real edge queue and synchronization tests.
14. Add real-time WebSocket updates.
15. Add restricted ANPR/face stages only if approved and justified.
16. Add model registry, evaluation, shadow mode, rollback, and drift monitoring.
17. Add production security hardening, observability, backups, recovery, and runbooks.
18. Rehearse the complete SIH scenario and document exactly what is simulated and what is measured.

21. Jury-safe summary of the complete feature set

IBVAP is a software-defined platform that keeps existing compatible CCTV as the observation layer and adds edge video processing, person and vehicle detection, temporary tracking, quality-aware rules, virtual-fence monitoring, night and direction context, incident correlation, alert prioritization, Behavioral Link Analysis, Evidence-Gap Intelligence, camera-health intelligence, secure evidence lineage,

offline synchronization, and authorized human review. Its key differentiator is that it does not hide uncertainty. It shows what was directly observed, what was inferred, what the camera could not verify, and what the human reviewer decided.

The professional claim is not that every production feature is already implemented. The professional claim is that the platform is decomposed into a measurable, testable architecture with a controlled prototype path and clear production gates.

References

- [1] ONVIF Profile M — Metadata and Events
- [2] ByteTrack: Multi-Object Tracking by Associating Every Detection Box
- [3] FastAPI WebSockets Documentation
- [4] NIST AI Risk Management Framework
- [5] NIST Face Recognition Technology Evaluation Program